

# Jonathan's Analytics Engineering Human Oversight Checklist

This checklist summarizes the key human interventions, architectural insights, and analytics engineering good practices identified during the DAI Ministry MEL/MIS Ecosystem Simulation project.

## 1. Validate Business Realism — Never Trust Simulated Data Blindly

- Validate simulations against real-world institutional scale and operational context.
  - Ensure analytical outputs make practical business sense.
  - Check proportionality between ministries, provinces, programs, and indicators.
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## 2. Validate Relationships — Never Assume Auto-Relationships Are Correct

- Manually review relationships created by Power BI.
  - Verify filter propagation behavior.
  - Keep cardinality clean and understandable.
  - Remove unused dimensions from the active semantic model.
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## 3. Understand Conformed Dimensions Properly

- Shared dimensions across facts are expected in enterprise analytics.
  - A visually complex schema can still be architecturally correct.
  - Focus on semantic correctness rather than visual neatness.
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## 4. Use Proper Date Dimensions for Time Intelligence

- Use a dedicated dim\_date table.
  - Include date\_key, date, year, month, quarter, and year\_month.
  - Connect all fact tables to the date dimension.
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## 5. Maintain Naming Consistency

- Use consistent naming conventions across dimensions and facts.

- Prefer readable enterprise-style naming such as date\_key.
  - Avoid unnecessary semantic ambiguity.
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## **6. Validate Conditional Formatting Logic Carefully**

- Always test threshold logic visually.
  - Remember Power BI percentages are stored as decimals.
  - Validate KPI coloring against actual business rules.
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## **7. Handle Data Quality Logic at the Database Layer**

- Move complex DQ rules into SQL/dbt models.
  - Keep Power BI visuals simple.
  - Use exception tables/views for operational monitoring.
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## **8. Validate Interactivity and Filter Context**

- Test slicers and cross-filter behavior thoroughly.
  - Most interactivity problems originate from semantic model design.
  - Confirm visuals respond correctly to dimension filters.
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## **9. Understand the Role of Measures**

- Measures operate across the semantic model through relationships.
  - Centralize reusable calculations in a dedicated measures table.
  - Avoid duplicated logic across visuals.
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## **10. Use AI as an Accelerator — Not as a Replacement for Thinking**

- Use AI for scaffolding, refactoring, and repetitive work.
  - Use human reasoning for architecture, semantics, and realism.
  - Validate all AI-generated outputs carefully.
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## 11. Understand That Enterprise Models Evolve Iteratively

- Enterprise analytics models improve gradually over time.
  - Refactor confidently as understanding deepens.
  - Continuous semantic improvement is normal.
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Final Reflection: The strongest interventions in this project were not coding interventions. They were semantic, architectural, business realism, and dimensional modeling interventions — the true foundation of analytics engineering.